

20.1 Magnetic Fields

Question Paper

Course	CIE A Level Physics
Section	20. Magnetic Fields
Topic	20.1 Magnetic Fields
Difficulty	Medium

Time allowed: 40

Score: /31

Percentage: /100

Question 1

A circular coil P carrying an alternating current produces a changing magnetic field. When a second similar coil Q is placed with its centre a distance x from the centre of coil P, as shown in Fig. 1.1, an electromotive force (e.m.f.) E is induced in coil Q.

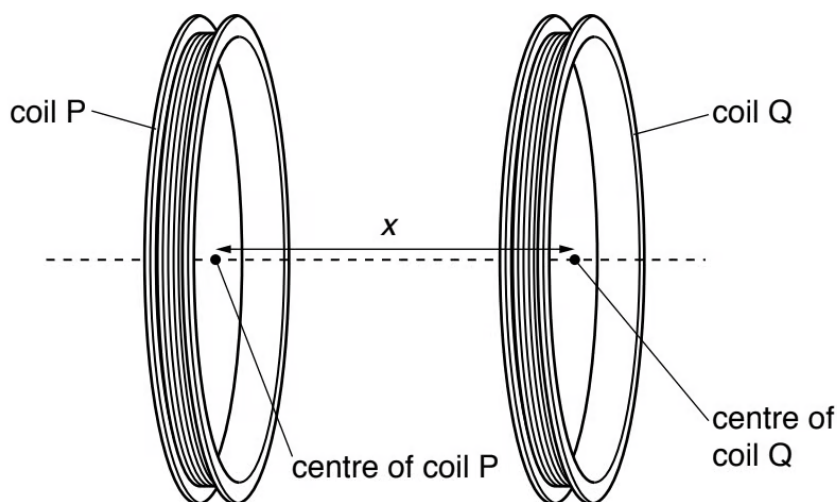


Fig. 1.1

It is suggested that E is related to x by the relationship

$$E = IZe^{-kx}$$

where I is the current in coil P, and k and Z are constants.

Plan a laboratory experiment to test the relationship between E and x .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for k and Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

[15 marks]

Question 2a

A thin slice of conducting material has its faces PQRS and VWXY normal to a uniform magnetic field of flux density B , as shown in Fig. 6.1.

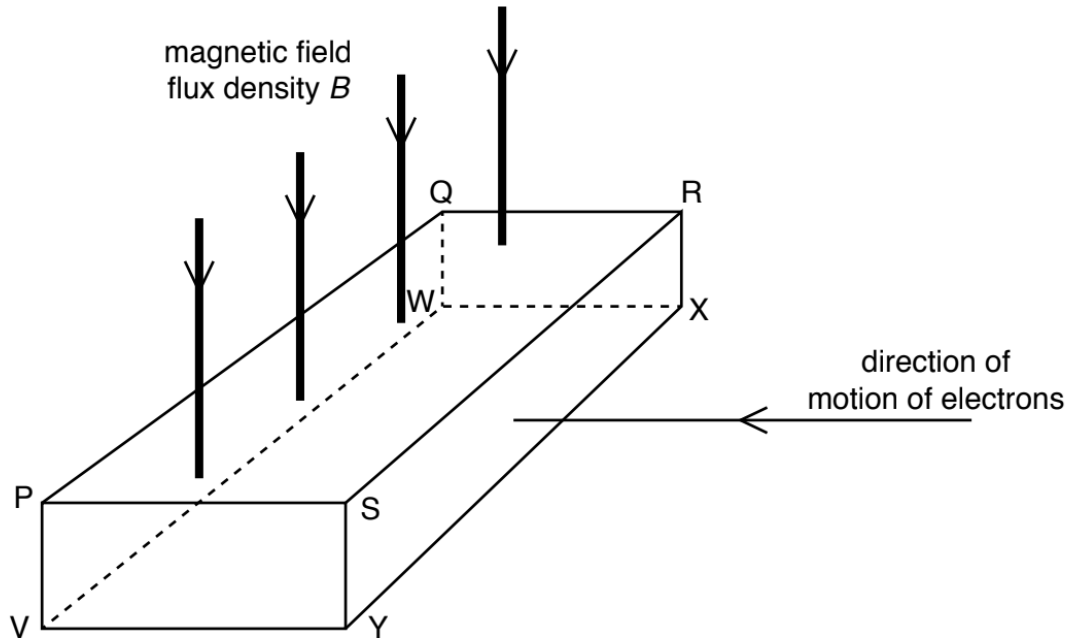


Fig. 6.1

Electrons enter the slice at right angles to face SRXY.

A potential difference, the Hall voltage V_H , is produced between two faces of the slice.

- (i)
Use letters from Fig. 6.1 to identify the two faces between which the Hall voltage is produced.

..... and [1]

- (ii)
State and explain which of the two faces named in (a)(i) is the more positive.

[2]

[3 marks]

Question 2b

The Hall voltage V_H is given by the expression

$$V_H = \frac{BI}{ntq}$$

(i)

Use the letters in Fig. 6.1 to identify the distance t .

[1]

(ii)

The negative charge carriers (electrons) are replaced by positive charge carriers moving in the same direction towards the slice.

State and explain the effect, if any, of this change on the polarity of the Hall voltage.

[2]

[3 marks]

Question 3a

State and describe the conditions that must be satisfied for a copper wire, placed in a magnetic field, to experience a magnetic force.

[2 marks]

Question 3b

Two long parallel current-carrying wires are placed near to each other in a vacuum.

Explain why these wires exert a magnetic force on each other. You may draw a labelled diagram if you wish.

[3 marks]

Question 3c

A long air-cored is connected to a power supply so that the solenoid creates a magnetic field. Fig. 1.1 shows a cross-section through the middle of the solenoid.

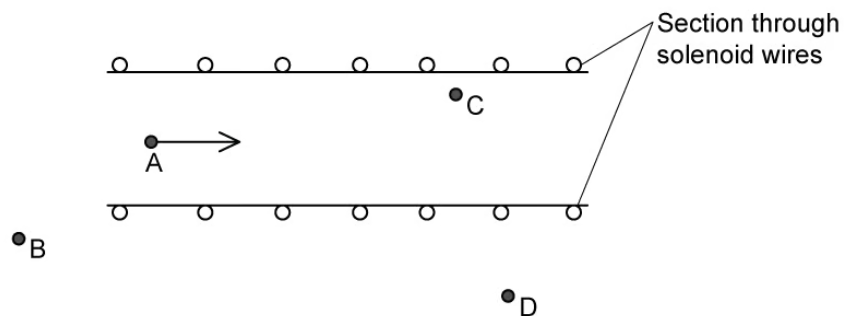


Fig. 1.1

The direction of the magnetic field at point A is indicated by the arrow. The other points are labelled B, C and D.

On Fig. 1.1, draw arrows to indicate the direction of the magnetic field at each of the points B, C and D.

[3 marks]

Question 3d

Compare the magnitude of the flux density of the magnetic field:

(i)

At A and C

[1]

(ii)

At A and D.

[1]

[2 marks]

Model Answers: Medium

1

Plan a laboratory experiment to test the relationship between E and x . Draw a diagram showing the arrangement of your equipment. Explain how the results could be used to determine values for k and Z .

Each point from the list can score **one mark** up to a **maximum of 15 marks** (*but read the guidance below in blue*):

Defining the problem:

- x is the independent variable and E is the dependent variable **OR** vary x and measure E ; [1 mark]
- The r.m.s current I_{rms} / peak current I_0 in coil P is the control variable; [1 mark]

Methods of data collection:

- A diagram of the apparatus that includes labels for coils P, Q and distance x **AND** coils P and Q supported by a stand; [1 mark]
- Circuit diagram 1: an a.c. power supply / signal generator connected to coil P in series with an ammeter **AND**
- Circuit diagram 2: a voltmeter / CRO connected to coil Q; [1 mark]
- A method to determine x e.g. use a ruler to measure the distance x between the coils; [1 mark]
- A method to measure x from the centre of the two coils e.g. measure the width of each coil and divide by 2, then add to the separation of the coils; [1 mark]

Method of analysis:

- Plot a graph of $\ln E$ against x ; [1 mark]
- The gradient of the graph is $-k$; [1 mark]

- The y-intercept = $\ln IZ$ **or** $Z = \frac{e^{y - \text{intercept}}}{I}$; [1 mark]

Additional detail including safety procedures:

Any **six** from:

- Do not touch the hot coil / use gloves to position the hot coil; [1 mark]
- Use a large current / a large number of turns / an iron core to produce a large magnetic field / induced e.m.f; [1 mark]
- Use a high frequency a.c. to produce a large induced e.m.f; [1 mark]
- Keep the number of turns on each coil constant / frequency of the a.c. constant; [1 mark]
- A method to check that the current is constant e.g. use a variable resistor to keep the ammeter reading constant; [1 mark]
- Take repeat measurements of x for different parts of the coil and take the average; [1 mark]
- A method to position the ruler horizontally to measure x as described e.g. use a spirit level; [1 mark]
- A method to keep the coils parallel e.g. adjust coil Q until the maximum reading is obtained; [1 mark]
- Taking logs of the given equation leads to $\ln E = -kx + \ln(IZ)$; [1 mark]
- This relationship is valid if a straight line is produced when the graph of $\ln E$ against x is plotted; [1 mark]

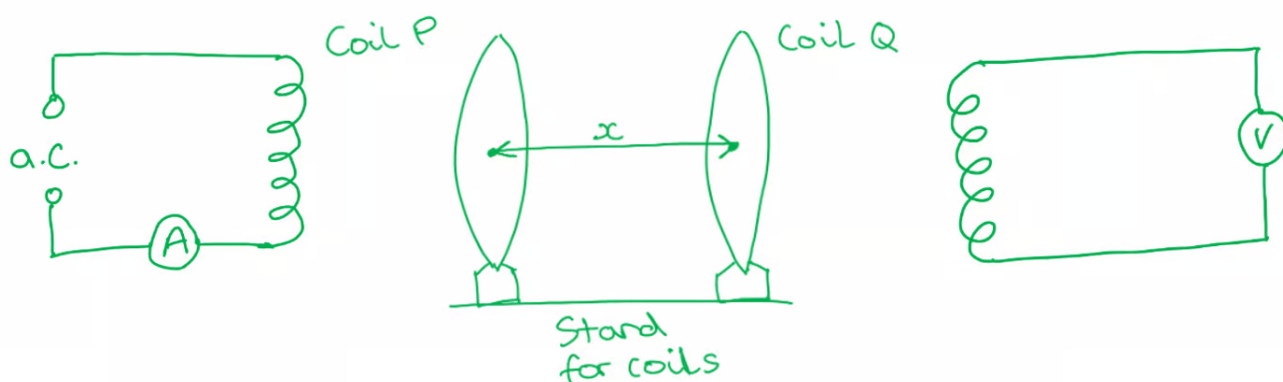
An example of an answer which would score 15 marks is:

Method

1. Set up the apparatus as shown in the diagram below [1] [2]. Use a ruler to measure the horizontal distance x between the two coils [3].
2. The distance x must be measured from the centre of coil P to the centre of coil Q. This centre can be located by measuring the width of both coils, dividing by 2 and then x can be obtained by adding these half widths to the separation of the coils [4]. A spirit level should be used with a ruler to ensure x is measured horizontally [5].
3. The independent variable is x and the dependent variable is E [6]. The current in coil P, the number of turns on each coil and the frequency of the a.c. must be

kept constant throughout [7].

4. For different values of x , repeat measurements of x should be taken at different points on the coil and then calculate an average [8].
5. Connect coil P to a high frequency a.c. to induce a large e.m.f which will in turn, generate a large magnetic field [9] and induce a current in coil Q.
6. To ensure the coils are parallel, adjust coil Q until the maximum reading is obtained [10]



Analysis:

1. Plot a graph using the mean values of $\ln E$ against x [11]
2. Taking logs of the equation $E = IZe^{-kx}$ leads to $\ln E = -kx + \ln IZ$ [12]. This relationship is valid if a straight line is produced when the graph of $\ln E$ against x is plotted [13].
3. Comparing this to the equation of a straight line, $y = mx + c$
 - Gradient = $-k$
 - y-intercept = $\ln IZ$
4. Determine a value for k using $k = -\text{gradient}$ and Z using $Z = \frac{e^{y - \text{intercept}}}{I}$ [14].

Safety considerations:

- The coils will get very hot, so gloves should be worn during handling [15]

[Total: 15 marks]

Question 1 in paper 5 assesses your **planning skills**. Your answer must comment on each of the following aspects for a maximum of **15 marks**:

- Defining the problem
- Methods of data collection
- Method of analysis

- Additional detail including safety considerations

Defining the problem:

- State the **independent**, **dependent** and **control variables** of the experiment

Methods of data collection:

- Describe the method used to change the independent variable
- Describe how the independent and dependent variables are measured
- Describe how the control variables are kept constant
- Use a **diagram** to describe the arrangement of the apparatus and the procedures to be followed

Method of analysis:

- Describe how the data should be used to reach a conclusion
- Include details of derived quantities derived from graphs plotted

Additional details including safety considerations:

Marks are available for additional information. How the marks are awarded will depend on the experiment. They could be for describing how additional variables can be kept constant, a circuit needed to make a particular measurement, a description of initial experiments.

Marks for safety considerations will be awarded for:

- Assessing the risks of the experiment
- Describing precautions that should be taken to keep risks to a minimum

- The experiment asks you to test the relationship between E and x .
- To determine the graph needed, rearrange the equation and match to $y = mx + c$

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Rearrange:

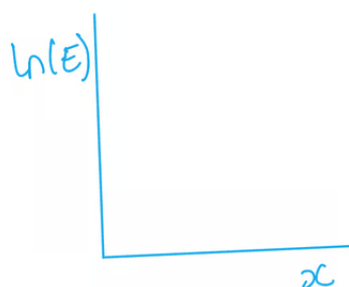
$$E = IZ e^{-kx}$$

Take $\ln$ of both sides:

$$\ln(E) = \ln(IZ) + \ln(e^{-kx})$$

$$\ln(E) = \underbrace{-kx}_{\downarrow y = mx + c} + \ln(IZ)$$

Hence the axis of the graph will be:



The gradient of the graph = $-k$

The y -intercept = $\ln(IZ)$

Rearrange the y -intercept to find an expression for Z :

Take natural exponentials of both sides:

$$e^{y\text{-intercept}} = e^{\ln(IZ)}$$

$$e^{y\text{-intercept}} = IZ$$

Divide both sides by I :

$$\frac{e^{y\text{-intercept}}}{I} = \frac{IZ}{I}$$

2a

(a)

(i) Consider the position of the magnetic field in relation to current:

- The Hall voltage states that the magnetic field is applied perpendicular to the current

Therefore the two faces between which the Hall voltage is produced are:

- PSVY and QRWX; [1 mark]

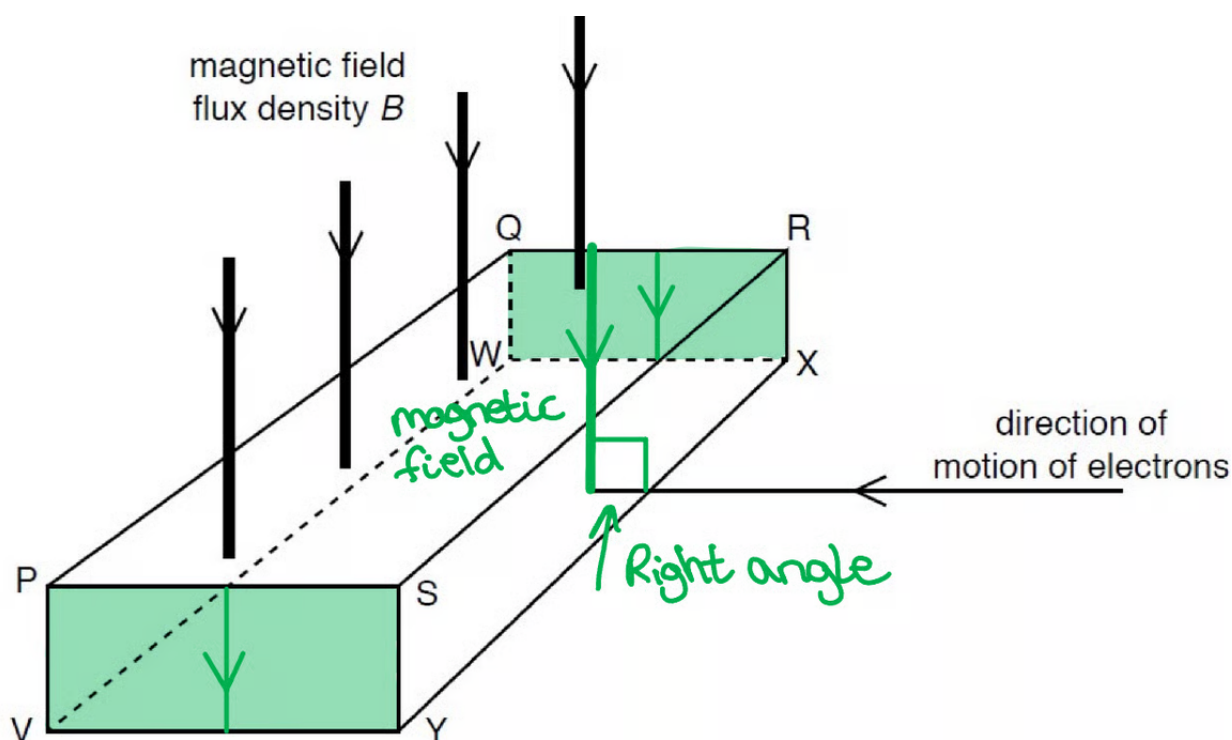


Fig. 6.1

(ii) Consider the deflection of an electron in the field:

- An electron moving in the magnetic field will be deflected towards face QRWX; [1 mark]
- Hence, the negative electrons will build up and create an overall surplus negative charge
- (On the other face PSVY, there will be a surplus of positive charge) so PSVY is

more positive; [1 mark]

[Total: 3 marks]

- In part (i) **draw** lines on the **diagram** to identify the faces perpendicular to the current and parallel to the magnetic field
 - Don't be fooled by **PQRS** this face is **parallel** to the current
- In part (ii) consider Fleming's Left-Hand Rule:
 - The magnetic field is represented by the first finger and acts down the page
 - The current is represented by the second finger and acts across the page to the right in the opposite direction to the electron flow
 - The thumb represents the subsequent motion of the electrons which will move into the page and toward face QRWX

2b

(b)

(i) Thickness t is equal to:

- PV **or** SY **or** RX **or** QW; [1 mark]

(ii) State and explain the effect, if any, of this change on the polarity of the Hall voltage:

- The negative and positive charges would deflect in opposite directions; [1 mark]
- So there is no change in the polarity (of the Hall voltage); [1 mark]

[Total: 3 marks]

- Part (i) requires you to consider the length through which the Hall voltage acts:
 - t stands for the **thickness** of the **conductor**
 - The **Hall voltage** acts between PSYV and QRXW, so the distance PQ is the **length** of the block
 - So, the thickness of the conductor is as **thick** as the block:
 - PV / SY / RX / QW
- Part (ii) is a little difficult to understand:
 - Using the same convention as part a, ii, under the influence of the **same magnetic field**

charge

- (On the other face PSVY, there will be a surplus of positive charge) so PSVY is more positive; [1 mark]

[Total: 3 marks]

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 - So, the thickness of the conductor is as **thick** as the block:
 - PV / SY / RX / QW
- Part (ii) is a little difficult to understand:

- Using the same convention as part a, ii, under the influence of the **same magnetic field**
- The **positive charge** carriers would move in the **opposite direction** to that of the electrons
- So the positive and negative charges will still **build up on the same sides**, as with the electrons
- So, the **electric field** remains in the **same direction**

3a

(a) The conditions that must be satisfied for a copper wire, placed in a magnetic field, to experience a magnetic force are:

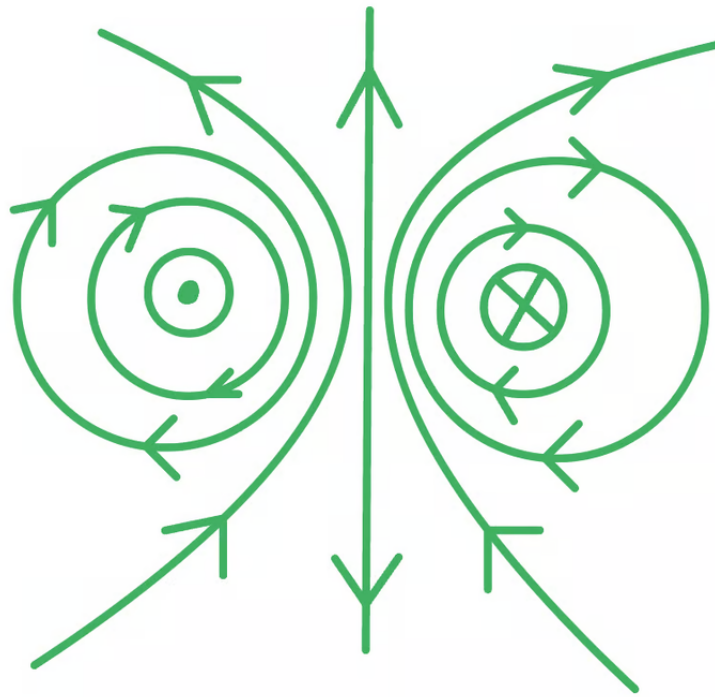
- Current: There must be current in the wire; [1 mark]
- Angle: The wire must be at a non-zero angle / not parallel to the magnetic field; [1 mark]

[Total: 2 marks]

3b

(b) Explain why the wires exert a magnetic force on each other:

Draw two parallel current-carrying wires with current travelling in the opposite directions:

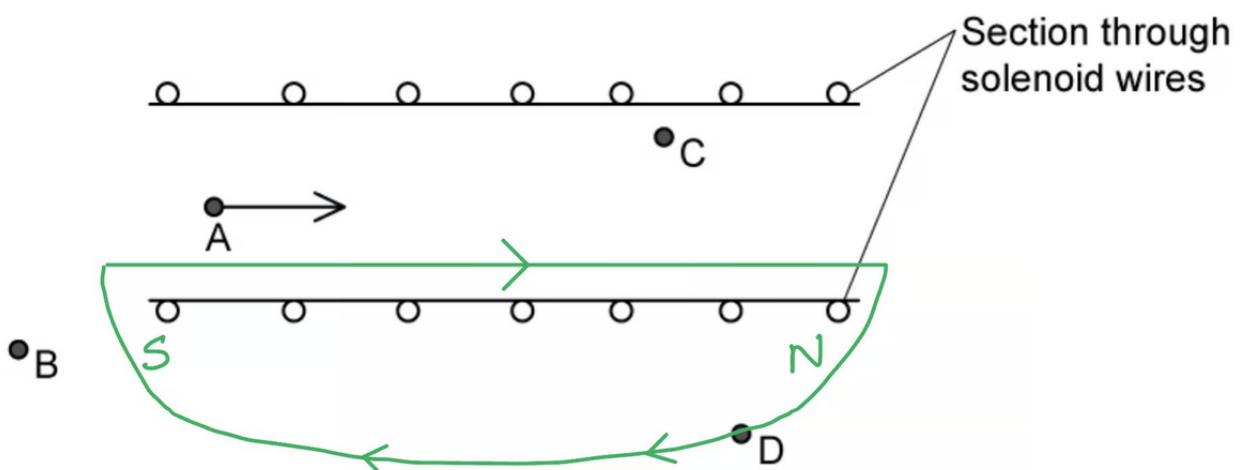


- The current in the wire creates a magnetic field around the wire; [1 mark]
- Each wire is present in the magnetic field created by the other; [1 mark]
- For each wire, the current / wire is perpendicular to the magnetic field due to the other wire, so it experiences a magnetic force; [1 mark]

[Total: 3 marks]

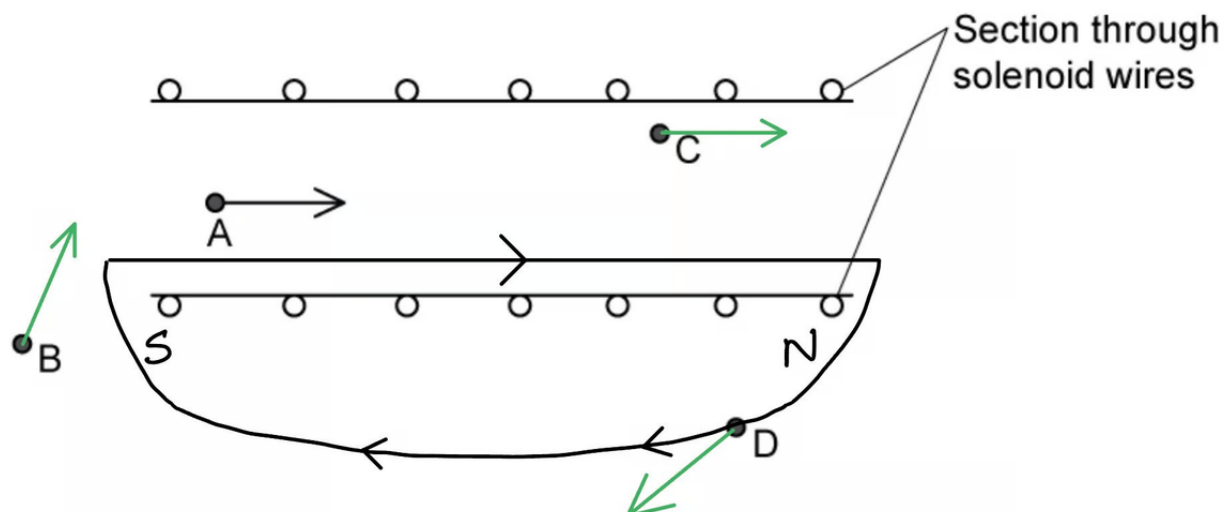
3c

(c) Draw a field line on the diagram to indicate the direction of the magnetic field within the solenoid:



- The direction of a magnetic field goes from north to south
- For the field to go from A to the right
 - The south pole must be on the left side
 - The north pole on the right side

Draw arrows on the diagram to indicate the direction of the magnetic field at B, C and D:



- Arrow from B pointing diagonally upwards and to the right at about 45° ; [1 mark]
- Arrow from C pointing horizontally to the right; [1 mark]
- Arrow from D pointing diagonally downwards and to the left at about 45° ; [1 mark]

[Total: 3 marks]

Draw the **arrows parallel** to the **field line** to ensure their direction is correct.

3d

(d)

Draw lines on the diagram to observe the density of the magnetic field lines at each point:

(i) The magnitude of the flux density of the magnetic field at A and C is:

- Equal; [1 mark]